No. Special requirement Related hazard
A-15 An appropriate procedure should be established for warm -up, inert gas purge, gas -free, hydrogen purge and pre -cooling.
The procedure should include:
.1 selection of inert gas in relation to temperature limit;
.2 measurement of gas concentration;

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The procedure should include:
.1 selection of inert gas in relation to temperature limit;
.2 measurement of gas concentration;
.3 measurement of temperature;
.4 rates of supply of gases;
.5 conditions for commencement, suspension, resuming and termination of each operation;
.6 treatment of return gases; and
.7 discharge of gases.
Prevention of dangerous purging operation
(see 4.11)
A-16 Only almost pure para-hydrogen (i.e. more than 95%) should be loaded in order to avoid excessive heating by ortho - to para-hydrogen conversion.
General
(see 4.1)
A-17 Fire detectors for detecting hydrogen fire should be selected, taking into account the features of hydrogen fire, to the satisfaction of the Administration.
Features of hydrogen fire and fire hazard
(see 4.7.4)
A-18 At the design stage, dispersion of hydrogen from vent outlets should be analysed in order to minimize risk of ingress of flammable gas into accommodation spaces, service spaces,
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should be analysed in order to minimize risk of ingress of flammable gas into accommodation spaces, service spaces, machinery spaces and control stations. Extension of hazardous areas should be considered based on the results of the analysis.
Low density and high diffusivity
(see 4.5)
A-19 Due consideration should be given to appropriate safety measures to prevent formation of explosive mixture in the case of a leakage and permeation of hydrogen, including:

.1 installation of hydrogen detectors in order to detect a possible ground -level travel of low temperature hydrogen gas, and at high points in spaces where warm hydrogen gas can be trapped; and
.2 application of "best practice" for land -based liquid hydrogen storage taking into account appropriate guidance such as "Cryogenics Safety Manual Fourth Edition (1998)"8).
General
(see 4.1)
A-20 In the case that fusible elements are used as a means of fire
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SOURCE: MSC.566(109).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.566(109).pdf
Edition (1998)"8).
General
(see 4.1)
A-20 In the case that fusible elements are used as a means of fire detection required by paragraph 18.10.3.2 of the Code, flame detectors suitable for hydrogen flames should be provided in addition at the same locations. Appropriate means should be adopted to prevent the activation of ESD system owing to false alarm of flame detectors, e.g. avoiding activation of ESD system by single sensor (voting method).
Fire hazard
(see 4.7.4)

No. Special requirement Related hazard
A-21 Consideration should be given to enhance the ventilation capacity of the enclosed spaces subject to liquefied hydrogen leakage and permeation, taking into account the latent heat of vaporization, specific heat and the volume of hydrogen gas in
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leakage and permeation, taking into account the latent heat of vaporization, specific heat and the volume of hydrogen gas in relation to temperature and heat capacity of adjacent spaces.
Low density and
high diffusivity
(see 4.5)
A-22 Liquid and gas hydrogen pipes should not pass through enclosed spaces in addition to other than those referred to in paragraph 5.2.2.1.2 of the Code, unless:
.1.1 the spaces are equipped with gas detection systems

which activate the alarm at not more than 20% LFL and shut down the isolation valves, as appropriate, at not more than 40% LFL (see sections 16.4.2 and 16.4.8 of the Code); and
.1.2 the spaces are adequately ventilated; or
.2 the spaces are maintained in an inert condition.
Susceptibility to leakage (see 4.4)
A-23 A risk assessment should be conducted to ensure that risks arising from liquefied hydrogen cargo affecting persons on board, the environment, the structural strength or the integrity of the ship
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arising from liquefied hydrogen cargo affecting persons on board, the environment, the structural strength or the integrity of the ship
are addressed. Consideration should be given to the hazards associated with properties of liquefied hydrogen and hydrogen gas, physical layout, operation and maintenance, following any reasonably foreseeable failure. For the risk assessment, appropriate methods, e.g. HAZID, HAZOP, FMEA/FMECA, what-if analysis, etc., should be adopted taking into account IEC/ISO 31010:2019 "Risk management Risk assessment techniques"7) and SAE ARP 5580 -2001 "Recommended failure modes and effects analysis (FMEA) practices for non-automobile applications"9).
General (see 4.1)
A-24 Relief valve sizing should be undertaken for the most onerous scenario. The evaluation should include the fire scenario and should consider the resulting magnitude of the heat flux on the cargo containment system.
High pressure hazard (see 4.8)

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should consider the resulting magnitude of the heat flux on the cargo containment system.
High pressure hazard (see 4.8)
A-25 A filling limit exceeding 98% at reference temperature should not be permitted.
High pressure hazard (see 4.8)
A-26 Bolted flange connections of hydrogen piping should be avoided where welded connections are feasible.

Susceptibility to
leakage (see
4.4.2)
A-27 Due consideration should be given to the invisible nature of
hydrogen fire from the viewpoint of safety of ships and especially
personnel in case of fire.
Fire hazard
(see 4.7.1)
3 EXPLANATION ON GENERAL REQUIREMENTS
3.1 Properties of liquefied hydrogen
The application of general requirements in the Code for liquefied hydrogen has been considered based on a comparison study on the physical properties of liquefied hydrogen and LNG. LNG and liquefied hydrogen are cryogenic liquids, non -toxic, and generate
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and LNG. LNG and liquefied hydrogen are cryogenic liquids, non -toxic, and generate
flammable high pressure gas. For reference, table 3 shows the comparison of physical
properties of hydrogen and methane, the major component of LNG.

Table 3: Comparison of physical properties of Hydrogen and Methane
Hydrogen Methane References
Boiling temperature (K)* 20.3 111.6 ISO1), Annex A, Table A.3
Liquid density (kg/m3)* 70.8 422.5 ISO1), Annex A, Table A.3
Gas density (kg/m3)** (Air: 1.198) 0.084 0.668 NIST RefProp10)
Viscosity (g/cms x 10-6)
Gas
Liquid
8.8
13.49